higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 2

13 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 True
1.2 False
1.3 False
1.4 False
1.5 True

(5 × 1) **[5]**

QUESTION 2

2.1 2.1.1 Sasol (1)

2.1.2 The reason behind the formation of Sasol was that since the country does not have its own oil reserves, this put the country at risk of having oil supply cut off from political conflicts.✓ It was important for the country to have an alternative source of petroleum,✓ and Sasol formed with the aim of producing petroleum products from coal as the country does not have.✓ (3)

2.1.3
- Plastics
- Adhesives
- Petrol
- Diesel
- Paraffin
- Tar
- Lubricants
- Solvents
- Detergents (Any 3 × 1) (3)

2.1.4
- Coal is a black solid substance which is formed within the earth's crust from layers of organic matter (plant leaves) build up after millions of years.
- Crude oil is a dark flammable liquid, found in areas known as crude oil reservoirs under Earth's surface in porous rocks or in gaps. (2)

2.2 2.2.1 It is a resource that cannot be replaced once it is depleted or finished.

2.2.2 Eskom

2.2.3
- Carbon
- Hydrogen
- Oxygen
- Sulphur
- Nitrogen (Any4 × 1) (4)

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2.2.4
- Coal is used a fuel for heating purpose
- For the extraction of metals
- To generate electrical energy
- Production of petrochemical
- Production of fertilizers
- Production of explosives
- Production of petrol

Any (4 × 1) (4)

[19]

QUESTION 3

3.1 3.1.1 Turbidity occurs when water contains visible matter in suspension that is caused by colloidal matter present in water.

3.1.2 Scaling refers to solid crusts that are formed in boilers and other heating elements as a result of water that contains magnesium or calcium carbonates or sulphates.

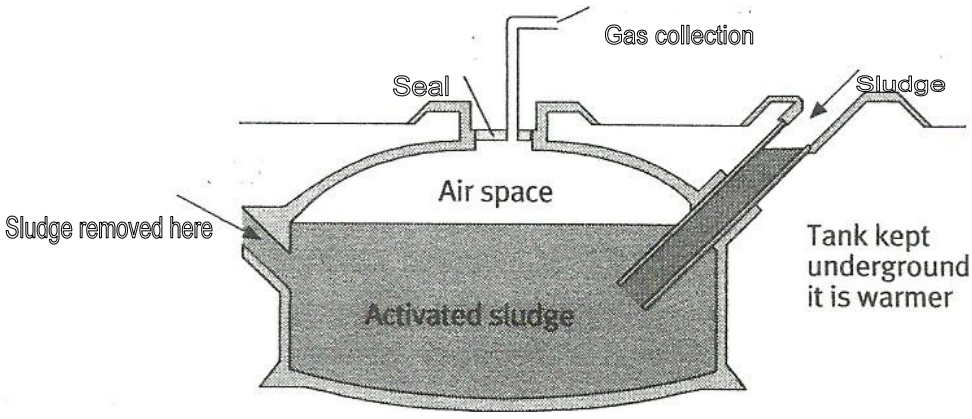
(2 × 2) (4)

3.2 Polluters must pay principle means that people or organisations that are responsible for polluting the environment must pay for all the damages caused as a result of pollution.✓ These include paying for the medical treatment of people or animals affected and ✓the repair of the damage caused by their activities.✓ (3)

3.3
- Pre-treatment
- Primary sedimentation
- Activated sludge treatment
- Disinfection
- Solid treatment

(5)

3.4



[1 mark for each correct labelling = 6]
[1 mark for correct diagram] (7)

3.5 Flash mixing is the rapid mixing of water✓ to make sure that the chemical flocculent added to water is thoroughly mixed with water.✓ The flocculent is mixed in as short time as possible in tank with a propeller turning at a high speed.✓ (3)

3.6
- Chlorine
- Ozone
- Chrome
- Iodine
- Heat
- Ultraviolet radiation
- Ultra filtration (Any 6 × 1) (6)

[28]

QUESTION 4

4.1 4.1.1 21% by volume (1)

4.1.2
- Colourless
- Odourless
- Tasteless
- Highly reactive
- Freezes at -218 °C
- Liquefies at -183 °C
- Boils at 183 °C (Any 4 × 1) (4)

4.1.3 78% by volume (1)

4.1.4
- Carbon dioxide
- Argon
- Neon
- Helium
- Krypton
- Xenon
- Water vapour
- Smog
- Nitrous oxide (Any 4 × 1) (4)

4.1.5
- Manufacture of ammonia
- Manufacture of explosives
- Liquid nitrogen is used as a cooling agent
- Used to provide an unreactive atmosphere when handling easily oxidisable chemicals
- Used as a carrier gas in vapour phase chromatography
- Used to preserve food (Any 5 × 1) (5)

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4.2 A – Water
B – Gas holders (impure C_2H_2)
C Low pressure drier
D High pressure Drier
E Acetylene (5)
[20]

QUESTION 5

5.1 5.1.1 Silica

5.1.2 Sodium sulphate

5.1.3 Feldspar

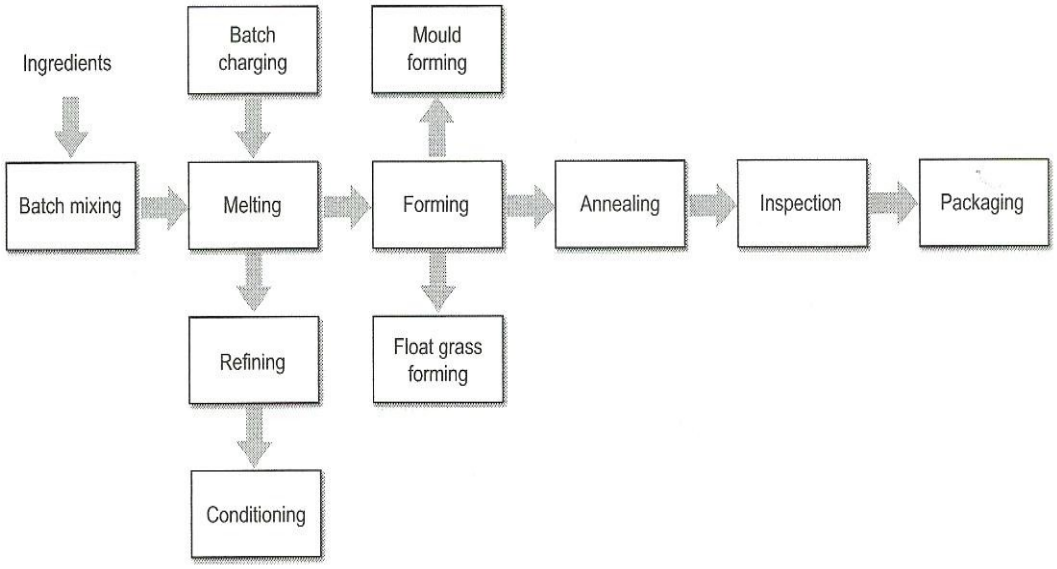
5.1.4 Cullet

5.1.5 Sodium carbonate

(5 × 1) (5)

5.2
- It is non-porous
- It is transparent and
- Virtually unreactive/inert (Any 2 × 1) (2)

5.3



(12)
[19]

QUESTION 6

6.1 6.1.1 To protect the wearer from inhaling dust particles and certain gases.

6.1.2 To give protection to human head against falling objects.

6.1.3 To protect the wearer against noise.

6.1.4 Protects the face of the person wearing when working with chemicals or any airborne particles or objects.

6.1.5 For full-body protection

(5 × 1) (5)

6.2 Eye wash stations provide emergency eye treatment for people exposed to hazardous materials.✓ Eye wash bowels are attached to emergency showers✓ and they provide a continuous flow of water and are recommended for workplace with hazardous materials.✓ Here you can rinse your eyes for the at least 10 minutes before you seek medical attention afterwards.✓ (4)

[9]

TOTAL: 100

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